1. Identity

Document	STAR-PKG-STAR-QUANTILE-EN · v2.1 · 2026-11-01
Package	star-quantile
Version	0.1.0
Capability	star.statistics.quantiles
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_quantile

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_quantile-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_quantile-0.1.0-py3-none-any.whl	68408d480a0e5cecd713e5247c3af14ae3f2456a96ed4707d3c3a27b998d122e
star_quantile.py	641b34712e9e935080c7bfa12e6618dacd740a18ad2bad9c080b830da6c0ea13
star_quantile-0.1.0-source.zip	aa79101a8c8f20521fee7fa96564787aed7f7be6b8d9c055004b85a0c89a7f6a

5. What it does, and its limits

Median, Hyndman-Fan type 7 quantile, interquartile range and median absolute deviation computed exactly in rational arithmetic and rounded once; on 500 datasets (2000 values) every value is identical to the exact rational value from SymPy, and NumPy, SciPy and CPython statistics agree within 7e-16 of the largest value.

One quantile definition out of the nine in use: results differ from packages that default to another one, and the module does not offer the others. The MAD is returned unscaled (multiply by 1.4826 to estimate a Gaussian standard deviation: that factor is an assumption about the data, not applied here). No weighted quantiles, no confidence intervals, no streaming estimate. "Exact" refers to the floats supplied.

- `quantile(values, q)` is definition 7 of Hyndman and Fan (linear interpolation between order statistics at position (n - 1) q), the default of R and NumPy; `median` is the 0.5 quantile; `iqr` is the 0.75 quantile minus the 0.25 quantile; `mad` is the median of the absolute deviations from the median, not scaled.
- Each result is the float nearest to the exact value of its definition on the numbers given (q included, as the exact value of its float); the order of the values does not matter.
- Measured on 500 datasets of 1 to 60 values (2000 values): every value identical to the exact rational value (SymPy) rounded once; NumPy, SciPy and CPython's `statistics` within 7e-16 of the largest |value|.
- Every function returns a float or raises `ValueError`: data that are not a list or tuple of 1 to 100000 finite numbers within 1e150 (booleans refused), q outside [0, 1] or not a finite number.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_quantile_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-quantile}: Median, quantiles, IQR and MAD, correctly rounded},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_quantile/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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